**Unit:** 6 Unit 8: Quadratics **Name:** ____________

**Topic:** Solving by Quadratic Formula (Discriminant) **Date:** ____________

# Foundation (Jalapeno)

**Q1.** What is the formula for solving $ax^2 + bx + c = 0$?

(A) $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

(B) $x = \frac{-b \mp \sqrt{b^2+4ac}}{2a}$

(C) $x = \frac{-b \pm \sqrt{b^2-2ac}}{2a}$

(D) $x = \frac{-b \mp \sqrt{b^2-2ac}}{2a}$

(E) $x = \frac{-b \pm \sqrt{b^2+2ac}}{2a}$

**Q2.** What does the discriminant $b^2 - 4ac$ tell us about the solutions?

(A) Number of solutions

(B) Type of solutions

(C) Both A and B

(D) None of the above

(E) Only real solutions

**Q3.** If the discriminant is negative, what type of solutions will the equation have?

(A) Two real solutions

(B) One real solution

(C) No real solutions

(D) Infinite solutions

(E) Two complex solutions

**Q4.** Solve $x^2 + 4x + 4 = 0$ using the quadratic formula.

(A) $x = -2$

(B) $x = -2, x = 2$

(C) $x = 2$

(D) $x = -4$

(E) $x = 4$

**Q5.** If $b^2 - 4ac = 0$, how many solutions will the equation have?

(A) Two distinct real solutions

(B) One real solution

(C) No real solutions

(D) Infinite solutions

(E) Two complex solutions

**Q6.** What is the value of the discriminant for $x^2 - 3x - 4 = 0$?

(A) $(-3)^2 - 4(1)(-4) = 25$

(B) $(-3)^2 + 4(1)(-4) = 25$

(C) $(-3)^2 - 4(1)(4) = -7$

(D) $(-3)^2 + 4(1)(4) = 25$

(E) $(-3)^2 - 4(1)(-4) = -7$

**Q7.** Solve $2x^2 - 5x - 3 = 0$ using the quadratic formula.

(A) $x = \frac{5 \pm \sqrt{61}}{4}$

(B) $x = \frac{5 \pm \sqrt{49}}{4}$

(C) $x = \frac{5 \pm \sqrt{25}}{4}$

(D) $x = \frac{5 \pm \sqrt{9}}{4}$

(E) $x = \frac{5 \pm \sqrt{1}}{4}$

**Q8.** If the discriminant is positive, what type of solutions will the equation have?

(A) One real solution

(B) Two distinct real solutions

(C) No real solutions

(D) Infinite solutions

(E) Two complex solutions

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**Open Ended Questions (FRQ)**

**Q9.** Solve $x^2 + 2x + 1 = 0$ using the quadratic formula and interpret the discriminant.

**Q10.** Find the value of $c$ that makes the equation $x^2 - 4x + c = 0$ have exactly one real solution.

**Chef's Tips**



- Tip 1: Ensure students understand the quadratic formula and its components.
- Tip 2: Remind students to interpret the discriminant to determine the type and number of solutions.
- Tip 3: Encourage students to check their solutions by plugging them back into the original equation.